

Name: _____

Date: _____

PERCENTAGES, (PERCENTAGE OF AN AMOUNT)

LO: I can calculate a percentage of an amount using mental and written methods.

Finding a percentage of an amount

To find a percentage of an amount, divide by 100 to find 1%, then multiply by the percentage needed. Or use the multiplier method.

Multiplier: 35% of 80 = $0.35 \times 80 = 28$. The percentage as a decimal IS the multiplier.

Quick examples

●	10% of 70 = 7	divide by 10
●	25% of 60 = 15	divide by 4
●	15% of 80 = 12	$10\% + 5\% = 8 + 4$
●	10% of 70 = 70	that is 100%
●	25% of 60 = 20	divided by 3 not 4

Key idea

Percentage means per hundred. Divide by 100 to find 1%, then scale up.

$$\text{P\% of A} = \frac{\text{P}}{100} \times \text{A}$$

Convert percentage to decimal, then multiply

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - using 10%

Find 30% of 90

$$10\% \text{ of } 90 = 9$$

$$30\% = 3 \times 10\%$$

$$3 \times 9 = 27$$

$$= 27$$

Find 10% first, then multiply by 3 for 30%.

Example 2 - using the multiplier

Find 45% of 200

$$45\% \text{ as a decimal} = 0.45$$

$$0.45 \times 200$$

$$= 90$$

$$= 90$$

The multiplier method works for any percentage.

Example 3 - building percentages

Find 17.5% of 60

$$10\% = 6, 5\% = 3, 2.5\% = 1.5$$

$$17.5\% = 10\% + 5\% + 2.5\%$$

$$= 6 + 3 + 1.5 = 10.5$$

$$= 10.5$$

Break awkward percentages into easier parts.

Example 4 - percentage of money

Find 12% of £350

$$1\% \text{ of } 350 = 3.50$$

$$12\% = 12 \times 3.50$$

$$= £42$$

$$= £42$$

Find 1% by dividing by 100, then multiply up.

YOUR TURN – PRACTICE (deliberate practice)

1. Common percentages

Use mental methods to find each percentage.

- a) 50% of 84
- b) 25% of 120
- c) 10% of 65
- d) 75% of 200

2. Multiples of 10%

Find 10% first, then use it.

- a) 20% of 150
- b) 40% of 85
- c) 60% of 45
- d) 90% of 70

3. Using the multiplier

Use the decimal multiplier to calculate.

- a) 35% of 80
- b) 12% of 250
- c) 85% of 60
- d) 4% of 350

4. Percentage of money

Find each percentage amount.

- a) 15% of £80
- b) 8% of £450
- c) 17.5% of £200
- d) 33% of £600

5. Mixed percentage problems

Calculate each percentage.

- a) 5% of 320
- b) 72% of 50
- c) 2.5% of 400
- d) 65% of £180
- e) 11% of 900

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) 10% of 45 = 4.5 ☐

Correct answer: _____

Reason: _____

b) 10% means divide by 100 ☐

Correct answer: _____

Reason: _____

c) 35% of 80 = $0.35 \times 80 = 28$ ☐

Correct answer: _____

Reason: _____

d) 25% of 60 = $60 \div 25 = 2.4$ ☐

Correct answer: _____

Reason: _____

e) To find 1%, divide the amount by 100 ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A school has **1200 pupils**. **45%** are boys. Of the boys, **60%** play a sport. Of the girls, **55%** play a sport. How many pupils in total play a sport?

Expression: _____

Simplified: _____

b) A shop sells a jacket for **£85**. This is **32% more** than the shop paid for it. How much did the shop pay? If the shop sells **24 jackets**, what is the total profit?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

